| | | |
|---|---|---|
| 6(a)(i) | energy stored = area under graph | **C1** |
| | $= \frac{1}{2} \times 450 \times 10^{-6} \times 8.0 = 1.8 \times 10^{-3}$ J **or** 1.8 mJ | **A1** |
| 6(a)(ii) | $C = Q / V$ **or** $E = \frac{1}{2}CV^2$ | **C1** |
| | $C = (450 \times 10^{-6}) / 8.0$ **or** $(2 \times 1.8 \times 10^{-3}) / 8.0^2$<br>$= 5.6 \times 10^{-5}$ F | **A1** |
| 6(b)(i) | $V = V_0 \exp(- t / RC)$ **and** $\tau = RC$ | **C1** |
| | $V = V_0 \exp(- t / \tau)$<br>$V_0 = 8.0$ V, and at one time constant, $t = \tau$<br>$V / 8.0 = \exp(-\tau / \tau)$, so $\ln(V / 8.0) = -1.0$ **or** $-\ln(V / 8.0) = 1.0$ | **A1** |
| 6(b)(ii) | [$t$ read from graph at $-\ln(V / 8.0) = 1.0$]: $\tau = 3.2$ s | **A1** |
| 6(b)(iii) | $\tau = RC$ | **C1** |
| | $R = 3.2 / (5.6 \times 10^{-5})$<br>$= 5.7 \times 10^4\ \Omega$ | **A1** |